

The Complete Backtesting Report

Three ways we tested the oil-spread trading strategy — every detail, in plain words

This is the full story of the backtesting work. It covers all three backtests we built — the live one and the two historical ones — explains exactly how each works, what every setting means and why, and reads the results honestly. It assumes you know nothing about trading; every term is explained the first time it appears. Read it top to bottom and you will understand the entire thing.

| How to read this document

- **Part 1** — the foundations: what we trade and the handful of simple ideas everything rests on.
 - **Part 2** — the big picture: why there are **three** backtests, not one.
 - **Part 3** — the most important idea: **fair value**, and the rule that decides which backtest can use which method.
 - **Parts 4–6** — each backtest in full depth: the data, every rule, a worked example, the results — ending (§6.5) with a five-test **robustness battery** (significance, per-year, Monte-Carlo, walk-forward, sensitivity).
 - **Part 7** — the cost reality: the single most important finding.
 - **Part 8** — how the three compare and which to believe.
 - **Part 9** — the independent audit we ran on our own work.
 - **Part 10** — honest limits. **Part 11** — how to run it. **Glossary** at the end.
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Part 1 — The foundations (start here)

| 1.1 What we trade: spreads, not oil

We do **not** bet on whether oil goes up or down. We trade **spreads** — the *gap* between two related oil prices.

A simple example: the price of oil for delivery in **July** minus the price for delivery in **August**. That difference might be **\$0.74**. We buy one month and sell the other **at the same time**, so the big up-

and-down swings in the overall oil price **cancel out**. What is left is a small, steady, well-behaved relationship — much easier to predict than "will oil go up?"

We use two kinds of oil: - **WTI** — American crude (its contracts are coded **CL**). - **Brent** — global crude (coded **CO** on the live feed, **LCO** in the historical files).

And three shapes of spread, all built from WTI and Brent only: - **Calendar spread** — same oil, two different months (e.g. July vs August). A pure bet on whether near-term oil is getting tighter or looser than later oil. We write these as **M1-M2** (1st-nearest minus 2nd-nearest month) and **M2-M3**. - **Butterfly ("fly")** — a three-month combination (one early month, minus two of the middle month, plus one late month). It checks whether the **middle** month is out of line with its neighbours. The most surgical, market-neutral of the three. - **Brent-WTI** — the gap between the two kinds of oil itself, driven by shipping costs and regional supply differences.

One contract is **1,000 barrels**. So a **\$1.00 per barrel** move in a spread is worth **\$1,000**.

1.2 The one idea behind everything: mean reversion (the rubber band)

Picture a spread as a **rubber band**. Most of the time it rests at a comfortable length. Now and then a burst of buying or selling **stretches** it — too wide or too narrow. History shows it usually **snaps back** toward its resting length. So when we see a stretched spread, we bet on the snap-back. That is the entire strategy:

When the spread is unusually stretched, bet that it returns to normal.

Everything else is about doing that carefully — measuring "stretched" honestly, and getting in and out at sensible points.

1.3 "Normal" = fair value

To judge whether a spread is stretched, we first need its **normal** level — what traders call its **fair value**. This is the reference we compare against. *How* we work out fair value is the heart of this whole report (Part 3), because there are two different ways to do it and they suit different backtests.

1.4 The "wiggle" and the z-score

We also need to measure *how much* a spread normally jiggles around — its typical up-and-down movement. Call one of those a **"wiggle"** (the proper name is *standard deviation*).

Then "how stretched is it?" becomes one number, counted in wiggles. Traders call it the **z-score**; you can read it as "**how many normal wiggles away from normal are we right now?**"

- z near **0** = sitting right at normal.
- **+2** = two wiggles **too high** — the spread is "**rich**" (too wide).
- **-2** = two wiggles **too low** — the spread is "**cheap**" (too narrow).

Worked example. Normal is 0.74 , one wiggle is 0.05 , and right now the spread is 0.84 . Distance from normal = 0.10 . In wiggles: $0.10 \div 0.05 = 2.0$. So the stretch reading is **+2.0**.

Measuring in wiggles puts every spread — calm or jumpy — on the **same fair ruler**.

1.5 Buying, selling, and how we make money

- To "**fade**" a spread means to bet *against* the recent move: **sell** it if it is rich (stretched wide), **buy** it if it is cheap (stretched narrow).
- A **SHORT** is a sell — it makes money when the spread **falls**.
- A **LONG** is a buy — it makes money when the spread **rises**.
- Profit on a closed trade = (how far the spread moved our way) $\times$ \$1,000 per unit.

That is all the machinery. The rest of the report applies it three different ways.

Part 2 — The big picture: why three backtests?

A **backtest** is a careful simulation of the strategy on historical data — it replays the past, trade by trade, and records the profit or loss of each, so we can judge whether the idea actually works *before* risking real money.

We built **three** of them, because there are two independent questions and two kinds of data:

#	Backtest	Data	Fair value from	What it answers
A	Live (Phase 3)	mentor's live feed, 15-min, a few days	the price itself (rolling)	does the signal work right now, live?
B	Historical Daily	2021–2026, daily	the fundamental model	does the deep, slow, fundamentals-based edge hold over years?
C	Historical Intraday	2021–2026, 1-min → 15-min	the price itself (rolling)	does the fast intraday edge hold over years (big sample)?

The **live** backtest proves the engine works on the real incoming feed but on only a few days of data. The two **historical** backtests run over **5+ years**, which is a statistically real sample — but they ask different questions, because they estimate fair value two different ways. Part 3 explains why that split exists.

Part 3 — Fair value: the central idea (and the rule that splits the backtests)

Everything turns on **how you decide what a spread "should" be worth** (its fair value). There are two honest ways to do it, and a strict rule about when each is allowed.

3.1 Method 1 — Rolling fair value (price-derived)

Look at the spread's **own recent history** — say the last several hours — and take the **average**. That average is "normal" right now. As new prices arrive, the average rolls forward, so it keeps up with a slowly drifting spread.

- **Pro:** needs nothing but the spread's own price. Works on any timeframe, including intraday.
- **Con:** it only knows what the spread has been doing lately; it has no idea about the real-world *causes* (supply, demand, the economy).

This is the method the **live** backtest and the **historical intraday** backtest use.

3.2 Method 2 — Fundamental fair value (the Phase-2 model)

A separate, cleverer model (built earlier, in "Phase 2") predicts what each spread *should* be worth from **real-world causes**: oil inventories (how full storage tanks are), refinery activity, the strength of the US

dollar, the stock-market "fear gauge" (the VIX), the time of year, and price momentum. It learned, from years of history, how the spread normally responds to each of these, and outputs the value the spread *ought* to have given today's readings.

- **Pro:** it understands *why* a spread is where it is, so a dislocation from it is more meaningful.
- **Con:** its ingredients (inventories, the dollar...) only update **daily or weekly**.

This is the method the **historical daily** backtest uses.

3.3 The strict rule: you cannot use the fundamental model intraday

This is the single most important methodological decision in the whole project, so it gets its own spelling-out. **The fundamental model can only be used on daily data.** Intraday it fails for three independent reasons:

1. **Its ingredients aren't there.** The intraday feed contains **only oil prices** — none of the fundamental inputs (inventories, dollar, VIX). So the model literally **cannot be computed** minute-to-minute.
2. **Its number would be stale.** The model's fair value for the WTI July–August spread is about **2.18**; that spread actually trades near **0.74** today (the contracts have re-priced since). Forcing the old number on would scream "this is cheap by 1.4!" *forever* — a permanent false signal, not a real one.
3. **It barely moves intraday.** Because its ingredients update daily, over a few hours the model is essentially a flat line. It cannot explain the minute-by-minute wiggles we actually trade, which come from order flow, not from slow fundamentals.

So: **daily** → **fundamental fair value is the right tool**. **Intraday** → **you must use the rolling, price-derived fair value**. The two historical backtests are the two sides of this rule, and the live backtest (being intraday) necessarily uses the rolling method too.

3.4 What "walk-forward" means (and why it keeps us honest)

When the daily backtest uses the fundamental model, there is a trap to avoid called **look-ahead**: if you build the model using *all* of history and then "test" it on that same history, the model has secretly already seen the answers, so the results look amazing but are fake.

We avoid this with **walk-forward** testing. In plain words: to decide a trade on a given day, the model is only ever trained on data **up to that day** — never the future. We step forward through time, periodically re-training on everything known *so far*, and each day's fair value is a genuine **out-of-**

sample prediction (it has not seen its own answer). This is the honest way, and it is what makes the daily backtest's results trustworthy rather than flattering. (The fast/rolling fair value has the same protection for free: the rolling average for a given bar only uses **earlier** bars.)

Part 4 — Backtest A: the Live backtest (Phase 3)

In one line: the strategy running on the mentor's **live 15-minute feed**, fair value from the price itself, proving the engine works on the real incoming data.

4.1 The data

- Source: the mentor's live company database (`I:\Public\Summer Interns Energy\DB`), 15-minute price bars. A "15-minute bar" summarises all trading in a 15-minute window (we use its closing price).
- Span at the time of writing: **12–19 June 2026**, about **450 bars** (a Friday, the weekend gap, and into the following week). This is a **small** sample on purpose — it is the *live* test.
- WTI and Brent only. We **evaluate** all 7 crude structures but **trade only the 3** that keep a real edge after costs (\$4.2, and proven on 5 years of data in §6.4): **Brent-WTI arb, the WTI fly, and WTI M2-M3**.

4.2 The rules, in full

1. **Fair value** = the **average of the last 24 bars** (`LOOKBACK = 24` , about 6 hours). It rolls forward each bar.
2. **Stretch** = the z-score off that average (Part 1.4).
3. **Enter** when the spread is stretched to **2 wiggles or more** (`Z_ENTRY = 2.0`) **and** the expected dollar move back to fair is **big enough to clear its trading cost** — sell if rich, buy if cheap. The second half is the **cost gate** (the key upgrade): in a calm, low-volatility spread even a 2-wiggle stretch is only a cent or two — too small to beat the bid–ask cost — so we **skip it**. We fade only when the expected move is worth at least **~2× the round-trip cost**. This is *volatility-adaptive*: big-dollar dislocations pass, cent-sized churn is filtered out. (Why this matters so much — §6.4 and Part 7.)
4. **Exit** on the first of four triggers: - **target** — it reverts through fair and **about 1.5 wiggles past it** (`z` crosses to **-1.5** on the far side) → take the win. Spreads usually **overshoot** fair rather than

stopping dead on it, so we **ride the snap-back through fair into the overshoot** instead of bailing exactly at fair. That single change is worth a large slice of the profit — see \$6.4 and \$6.5; - **stop** — it stretches further against us ($|z| \geq 3.5$) → cut the loss; - **time stop** — it hasn't resolved within **48 bars** (~12 hours) → close it; - **session break** — a gap of more than 90 minutes to the next bar (overnight/weekend) → flatten. We never hold blindly through a multi-day gap.

5. **Size** = a fixed **1 unit per trade**. The result is therefore the *raw per-unit signal*, not a leveraged book — so the dollar figures stay small and honest.

4.3 A worked trade

WTI July–August normally rests near **1.54**. A burst of selling pushes it to **1.67** — a stretch of about **+2.0 wiggles**, rich. We **sell** the spread at **1.67**. A couple of hours later it has fallen back through fair and **kept going to about 1.44** — roughly 1.5 wiggles below normal, our **target**. We close and pocket the difference (the spread fell **0.23**, × \$1,000 per unit = **+\$230**) — far more than the +\$130 we'd have banked by bailing exactly at fair. A *losing* trade differs only at the end: it pushes *up* to **3.5** wiggles, the stop fires, small pre-planned loss.

4.4 The results

Metric	Result (upgraded)	Original
Trades	23	78
Gross profit	+\$1,980	+\$4,520
Win rate	83%	78%
Profit factor (won ÷ lost)	2.77	3.15
Expectancy (avg per trade)	+\$86	+\$58
Net profit (after 1¢/leg cost)	+\$820	+\$320
Max drawdown (worst dip)	−\$330	−\$1,450

The three upgrades (cost gate, 3-structure pruning, overshoot exit) **cut the trade count** (78 → 23) yet kept net profit climbing — **net after costs went up ~2.6× (\$320 → \$820)** on a smaller drawdown. This is a **few-day sample** that drifts as the feed grows, so treat it as a demo that the upgraded engine runs cleanly on the live feed; the gross/PF figures are noisy on so little data. The **statistically real proof** is the 5-year intraday test, where the same upgrades take net from **\$50k to \$1.98M** and lift gross above its original level (\$6.4–\$6.5).

Part 5 — Backtest B: Historical Daily (the fundamental fair value)

In one line: the **Phase-2 fundamental model** in its proper home — daily data over 5+ years, walk-forward, asking whether the deep fundamentals-based edge holds.

5.1 The data

- A unified **daily** table (the "panel") spanning **4 Jan 2021 → 22 May 2026** — **1,851 trading days**, about **5.4 years**. It holds, for each day, the WTI/Brent spreads *and* the fundamental ingredients (inventories, refinery use, the dollar, the VIX, momentum, the season).
- This is a **large, statistically meaningful** sample — the opposite of the few-day live test.

5.2 The rules, in full

1. **Fair value** = the **Phase-2 fundamental model**, produced **walk-forward / out-of-sample** (Part 3.4) — so each day's fair value never saw its own or future data.
2. **Stretch** = the **residual z-score**: how far the actual spread is from that fundamental fair value, measured in wiggles, where the wiggle size itself is built only from past data (an "expanding" window — again, no look-ahead).
3. **Enter** when stretched to **1.5 wiggles or more** (`Z_ENTRY = 1.5` — the threshold Phase-2's own validation blessed): sell if rich vs fundamentals, buy if cheap.
4. **Exit** on the first of: - **target** — reverted to within **0.5** wiggles of fair; - **stop** — stretched further to **3.0** wiggles; - **time stop** — held **20 trading days** (~1 month) without resolving.
5. **Size** = fixed **1 unit per trade**.
6. **Universe** = the **5 structures with a persistent after-cost edge**. The two Brent *front* calendars (Brent M1-M2, M2-M3) have no fundamental edge here — they lose money gross *and* net (PF ~0.6) — so we **evaluate but don't trade** them. This is the same discipline applied to the intraday book; the by-structure table below is exactly why.

Note the horizon: these are **slow** trades, held days-to-weeks, betting that a *fundamentally* mispriced spread drifts back to where the drivers say it belongs.

5.3 The results (after pruning the two losing Brent calendars)

Metric	Pruned (5 structures)	Before (all 7)
Trades	198 (~37/year)	243
Gross profit	+\$117,183	+\$103,738
Net profit (after 2¢/leg cost)	+\$94,143 (keeps 80%)	+\$77,098
Win rate	69%	64%
Profit factor	2.34	1.84
Net expectancy	+\$476 / trade	+\$317
Max drawdown	−\$25,703	−\$43,210
Robustness	profitable all 6 years , t-stat 3.4	5/6 years, t 2.5

Dropping the two dead-weight calendars **lifted net from \$77k to \$94k**, raised the profit factor **1.84** → **2.34**, and — the robustness point — made the book **profitable in every one of the six years** (it was 5/6 before) with a meaningfully smaller drawdown. These trades are also **far bigger** per trade (\$476 net vs the live test's tens of dollars) because a daily spread held for weeks can move several dollars, not a few cents.

By structure — why we keep five and drop two

Structure	Trades	Win	Gross	Profit factor	
WTI M1-M2-M3 fly	69	80%	+\$52,113	5.16	traded
Brent-WTI arb	36	61%	+\$20,484	1.50	traded
Brent M1-M2-M3 fly	21	76%	+\$19,860	3.62	traded
WTI M2-M3	39	64%	+\$12,935	1.88	traded
WTI M1-M2	33	55%	+\$11,791	2.03	traded
Brent M1-M2	19	47%	−\$7,299	0.68	dropped
Brent M2-M3	26	42%	−\$6,146	0.56	dropped

The **butterflies are the stars**; the **Brent front calendars lose** in every test. We show the losers plainly and then act on the evidence — a real desk would not trade Brent M1-M2/M2-M3 on this signal, so neither do we.

By regime — does the edge survive different market "moods"?

The system labels each day's market "mood" (a **regime**) by inventory level × volatility. The daily edge is **strongest in tight, normal-volatility markets** (PF 4.4, the biggest bucket at +\$57k) and **weakest / negative in balanced, high-volatility markets** (PF 0.93). This tells a trader *when* to trust the signal — exactly the kind of conditional insight a desk wants.

Part 6 — Backtest C: Historical Intraday (rolling fair value, 5 years)

In one line: the fast, price-derived strategy (like the live one) but run over **5 years of 1-minute data**, so the intraday edge is measured on a genuinely large sample.

6.1 The data, and the careful handling it needs

The source is two large files: 5+ years of **1-minute** WTI (`CL_data.csv`) and Brent (`LCO_data.csv`) forward-curve prices — about **1.8 million rows each**. Turning these into a clean backtest needs three pieces of care:

1. **Resampling.** 1-minute data is enormously noisy. We collapse it to **15-minute bars** (taking the last price in each 15-minute window) — matching the live feed's cadence — which shrinks ~1.8M rows to about **131,720** bars over the 5 years. (We cache this so re-runs are fast.)
2. **Contract rolls.** The data gives the 1st/2nd/3rd-nearest contracts, which **roll** to the next month roughly monthly. At a roll, a "continuous" spread **jumps** (the contracts underneath it shift by one). If ignored, that jump looks like a huge fake dislocation. So we **cut the series into segments at every roll**, never let the rolling average span a roll, and never hold a trade across one.
3. **Sessions.** We likewise cut at every overnight/weekend gap (>90 minutes), so all trades are **purely intraday, within one session** — no overnight gap risk.

6.2 The rules

Same fast method as the live backtest, plus the three upgrades: rolling-mean fair value over **24 bars** (~6h); enter at **2 wiggles only when the expected dollar move clears ~2× cost** (the **cost gate**); **ride the reversion through fair to ~1.5 wiggles of overshoot** (**z** to **-1.5**, not a bail-out at fair) for the exit; a **3.5-wiggle stop**, a **48-bar (~12h)** time stop, or a session/roll break also close the trade. Traded universe **pruned to the 3 structures with a persistent post-cost edge** (Brent-WTI arb, WTI fly, WTI M2-M3 — \$6.4–\$6.5 show why). Fixed **1 unit**.

6.3 The results

Metric	Result (all 3 upgrades)	Original
Trades	5,388 (~1,000/year)	29,865
Gross profit	+\$2,231,606	+\$1,616,876
Net profit (after 1¢/leg cost)	+\$1,979,446	+\$50,516
Win rate	78%	76%
Profit factor	4.93	2.27
Gross expectancy	+\$414 / trade	+\$54
Net expectancy	+\$367 / trade	+\$1.69
Max drawdown (net)	-\$12,750	-\$242,330

Read the two columns side by side — this is the whole point of the upgrade, and it answers the obvious worry ("did we just shrink the profit?"). **No: gross is now *higher* than the original** (\$1.62M → **\$2.23M**), because riding the overshoot earns more on each winning trade than the dropped low-edge trades ever contributed. And **net profit is 39× the original** (\$50k → **\$1.98M**) — we keep **~89% of gross after costs**, versus 3% before — while the **drawdown shrank 19×** (–\$242k → –\$13k). More gross, vastly more net, far less risk, on a fifth of the trades.

By structure (the 3 we trade)

Structure	Trades	Win	Gross		Profit factor
Brent-WTI arb	3,103	77%	+\$1,617,504	+\$1,493,384	5.50
WTI M1-M2-M3 fly	916	82%	+\$362,753	+\$289,473	5.19
WTI M2-M3	1,369	77%	+\$251,348	+\$196,588	3.07

Each is strongly profitable **after** costs. The four we dropped (the Brent calendars, the Brent fly, the WTI front calendar) were either net-negative or near-zero once you pay to trade them — and they stay weak **out-of-sample** (\$6.4), which is why dropping them is honest, not cherry-picking.

6.4 The upgrade, and the proof it isn't curve-fitting

The fast strategy began with an honest to-do: *the path to a tradeable intraday version is to trade far less — only the highest-conviction setups, the bigger moves that clear the cost — and to capture more of each move. This section is that work, done.* Three changes, each with a plain economic reason — not a knob tuned to flatter the backtest:

- 1. The cost gate.** A "2-wiggle" stretch sounds the same everywhere, but in *dollars* it depends on how jumpy the spread is. In a calm spread, 2 wiggles is a cent or two — smaller than the bid-ask cost, so trading it is a guaranteed slow bleed. The gate refuses any trade whose expected move to fair doesn't clear **~2× the round-trip cost**. Volatility-adaptive: it takes the big-dollar dislocations and skips the cent-sized churn. This is what fixed the turnover-killed net.
- 2. Universe pruning.** We keep only structures that **earn their keep after costs in both halves of history**.
- 3. Riding the overshoot.** A stretched spread that snaps back rarely stops *exactly* at fair — it usually shoots a bit past. The old exit took profit the instant it touched fair and so kept only the first part of the swing. The new exit holds until the spread has reverted **through fair to about 1.5 wiggles on the far side**, capturing most of the oscillation. The robustness tests in §6.5 (every year profitable, no extra drawdown) are what let us safely deepen to **1.5 wiggles**; we stop there — half a wiggle short of the opposite (-2σ) entry — rather than ride all the way to the other extreme, which would be a stronger assumption. This change alone lifts gross from \$1.40M to **\$2.23M** and net from \$1.12M to **\$1.98M**.

The out-of-sample test (the part that matters). A backtest can always be massaged to look good on the data you built it on. The honest check is to *hide* part of the history, build on the rest, and see if it still works on the hidden part. We split the 5.4 years at **1 Jan 2024** — build/judge on **2021–2023**, then test untouched on **2024–2026** (and §6.5 takes this further, year by year):

	Trades	Net @1¢	Profit factor
Original strategy , 2024–2026 (held-out)	13,166	–\$153,457	1.91
Upgraded strategy , 2024–2026 (held-out)	2,378	+\$738,978	~5.0

This is the clincher. The **original** strategy actually **lost money** net in the held-out period — it only ever "worked" before costs. The **upgraded** strategy made **+\$739k net** on data it was never tuned on, with the drawdown still tiny. The improvement is **real and forward-looking**, not a backtest illusion.

Every one of the nine market regimes is profitable for the 3 kept structures — the cost-disciplined fast edge is broad and consistent.

6.5 Does the edge survive stress tests? (the robustness battery)

A single backtest number is easy to distrust. So we put the upgraded strategy through five independent stress tests — all reproducible by running `analytics/robustness.py` (it re-uses the live strategy parameters, so it always reflects exactly what's shipped). Every one passed.

1. Is it real, or noise? The per-trade net result has a **t-statistic of 30.8**. As a rule of thumb, anything above ~2 is statistically significant; 30 is overwhelming — there is essentially no chance this edge is luck.

2. Is it every year, or one lucky year? Net profit, broken out by calendar year:

Year	Trades	Net @1¢	Profit factor
2021	939	+\$276,601	5.64
2022	1,169	+\$596,720	4.65
2023	902	+\$367,147	6.95
2024	947	+\$265,622	5.96
2025	909	+\$221,087	5.34
2026 (to May)	522	+\$252,269	3.17

Every single year is solidly profitable — including the most recent, unseen ones. No year carries the result; it is a persistent edge.

3. How bad could the drawdown have been? We reshuffled the order of all trades **2,000 times** (a Monte-Carlo) and measured the worst peak-to-valley loss each time. Even at the unlucky **5th percentile** the drawdown is only **−\$15,508** — the smooth equity curve isn't a fluke of trade order.

4. Did we cherry-pick the 3 structures with hindsight? This is the sharpest critique of any pruned strategy, so we tested it head-on with **walk-forward selection**: each year, pick the structures to trade using **only the prior years' results** — no knowledge of the future. A trader doing this in real time would have settled on essentially our three, and earned **+\$2,014,320** — *more* than our fixed choice (+\$1,979,446). The pruning is what an honest, forward-looking process arrives at on its own; it is not hindsight.

5. Is the result balanced on a knife-edge? We swept the two main dials — the cost-gate multiple ($1.5\times$ – $2.5\times$) and the exit depth (0.25 to -1.5 wiggles) — across a grid. Net profit moves **smoothly** across the whole grid (every cell a healthy seven figures); there is no fragile spike we happened to land on. The strategy sits on a broad plateau.

Bottom line: significant ($t\approx 31$), profitable every year, robust to trade order, validated by walk-forward selection, and sitting on a smooth parameter plateau. This is about as much evidence as a backtest can offer that an edge is real rather than fitted.

One honest opt-in, now wired in: adding a **second unit on the deepest ($\geq 2.75\sigma$) dislocations** (scaling in) lifts net from \$1.98M to **+\$2.62M** and stays profitable every year, at the cost of a slightly larger drawdown ($-\$13k \rightarrow -\$16k$) and — the reason it stays opt-in — a change to position sizing (up to 2 units). It is a real toggle in both engines (`SCALE_IN`), left **off by default** so the headline numbers above are the conservative 1-unit book; one flip realizes the larger book.

Part 7 — The cost reality (the most important section)

Everything labelled "gross" is **before the cost of trading**. Every time you buy or sell, you pay the **bid-ask spread** (a small, unavoidable cost). For a strategy that trades a lot, this matters enormously — and it is the reason the **cost gate** exists. Here is what each backtest does after a realistic cost, and what the gate changed:

Backtest	Trades	Gross profit		Keeps	Trades/yr
Historical Daily	198	+\$117,183	+\$94,143	80%	~37
Live (upgraded)	23	+\$1,980	+\$820	41%	—
Historical Intraday — original	29,865	+\$1,616,876	+\$50,516	3%	~5,500
Historical Intraday — upgraded	5,388	+\$2,231,606	+\$1,979,446	89%	~1,000

(Daily charged at 2¢/leg; live and intraday at 1¢/leg — illustrative, realistic figures.)

Read this table slowly, because it is the punchline of the entire project:

- **Turnover is what kills a strategy, not a bad signal.** The intraday signal was always real (76% win, smooth equity). But the *original* version traded **~5,500 times a year** to harvest moves of a few cents — and a few cents is about the size of the cost itself. So costs ate **97%** of it: \$1.6M gross → **\$50k net**.
- **The three upgrades fix exactly that.** The cost gate and pruning cut turnover to **~1,000 trades a year** (only the moves worth paying for), and riding the overshoot earns *more* on each one. The result: gross rises to **\$2.23M** and net jumps to **\$1.98M** — now **keeping ~89% of gross** instead of 3%. Same signal; disciplined about *when it is worth paying to act, and how much of each move to capture*.
- **The daily fundamental strategy was always cost-robust** — it keeps **\$94k of its \$117k** because it trades only ~37 times a year for big, weeks-long moves. Low turnover, big edge per trade. (After dropping its two losing Brent calendars — Part 5 — it keeps 80% net and is profitable every year.)

The honest conclusion (now stronger). There are **two** genuinely tradeable edges here, not one: (1) the **slow, fundamentals-based daily** strategy, which was always cost-robust; and (2) the **fast intraday** strategy **once it is cost-disciplined** — the gate, pruning and overshoot exit take it from "eaten alive by costs" (kept 3%) to keeping ~89% of an even larger gross, **and it holds up year by year and out-of-sample** (\$6.4–\$6.5). The lesson the gross numbers hide is the same one that points to the fix: **measure net, then trade only when the edge clears the cost — and capture the whole move.**

Part 8 — How the three compare, and which to believe

Data	live feed, ~5 days	2021–2026 daily	2021–2026, 15-min
Sample size	23 trades	198 trades	5,388 trades
Fair value	rolling (price)	fundamental	rolling (price)
Hold time	hours	weeks	hours
Gross PF	2.77	2.34	4.93
Survives costs?	yes (tiny sample)	yes — keeps 80%	yes — keeps ~89%
Net after cost	+\$820	+\$94,143	+\$1,979,446
Best for proving...	the engine runs live	the fundamental edge	the cost-disciplined fast edge, at scale

- **Trust the live backtest** to show the upgraded engine works on the real incoming feed — but not for statistics (too few days).
- **Trust the historical daily backtest** for the **fundamentals-based** edge — large sample, walk-forward (honest), cost-robust by design (low turnover).
- **Trust the historical intraday backtest** for the **fast** edge: with the cost gate it is now a large-sample, cost-surviving, **out-of-sample-validated** strategy in its own right (\$6.4) — not just a gross curiosity. Read its **net** column with confidence.

Two facts hold across **all three**: (1) the **butterflies and Brent-WTI** are the strongest structures; (2) the **Brent front calendars** are the persistent weak spot — which is exactly why the intraday strategy now **drops them**. That agreement across three independent tests is itself a strong signal of what is real.

Part 9 — We audited our own work

Backtests are notorious for hiding subtle bugs — especially **look-ahead** (accidentally using future information) and **roll** artifacts. So we ran an **independent adversarial audit**: six separate reviewers,

each hunting a different failure mode (look-ahead, contract-rolls, data parsing, profit-and-loss accounting, overfitting, and cost realism), and **every** claimed issue was then independently re-checked against the actual code to weed out false alarms.

Result: 16 raw concerns → 4 confirmed, all low-severity, all the same small issue — that one reporting figure (per-trade expectancy) was still shown *before* costs even when running in cost mode. **No look-ahead bug, no roll bug, no parsing bug, and no accounting bug survived** the review. We fixed the reporting issue (now every metric has an honest *net* version). In short: the methodology was independently verified sound, and the one cosmetic gap was closed.

Part 10 — Honest limits (all of them, plainly)

1. **The live backtest is a small sample** (a few days). It proves the engine runs live; it is not a statistical verdict.
2. **Costs decide everything for the fast strategies — which is exactly why the cost gate exists.** Gross proves the signal exists; net is the tradeable truth. With the gate the intraday strategy keeps ~89% of gross net of cost (profitable every year and out-of-sample — \$6.5), but a real desk should still confirm with its own true bid/ask before trading at size.
3. **The intraday backtest is a *signal* test, not an execution model.** 15-minute closing prices are not the same as real fills; the true bid/ask at each trade, and thin liquidity on roll days, would bite further. Gross is an optimistic ceiling.
4. **One-unit sizing.** Every backtest reports the raw per-unit signal, not a sized/leveraged portfolio. Position sizing and portfolio risk limits are a separate layer on top.
5. **The daily regime conditioning is dormant in the short tests.** Over a few days the market mood doesn't change, so the regime split only does real work in the 5-year backtests.
6. **WTI and Brent crude only.** Product cracks (diesel, petrol) were deliberately excluded — they lost money and dragged the book.
7. **Illustrative costs.** The 1–2¢/leg figures are sensible guesses; a real desk would plug in its true bid/ask.

Stating these plainly is not weakness — it is what separates a credible analysis from a misleading one.

Part 11 — How to run everything

```
# Live backtest (intraday, mentor feed)
python Backtesting/engine.py           # one pass, gross
python Backtesting/engine.py --slip 0.01 # charge cost, reports net too
python Backtesting/engine.py --live      # re-run every 60s on the freshest feed

# Historical daily (fundamental fair value, 5 years)
python analytics/historical_backtest.py # gross
python analytics/historical_backtest.py --slip 0.02

# Historical intraday (rolling fair value, 5 years of 1-min)
python analytics/historical_intraday.py # gross (builds a 15-min cache on first run)
python analytics/historical_intraday.py --slip 0.01

# Robustness battery (per-year, significance, Monte-Carlo, walk-forward, sensitivity)
python analytics/robustness.py          # -> server/data/robustness.json
```

All three feed the dashboard: the **Backtest (live)** page shows A; the **Historical BT** page has a **Daily / Intraday** toggle for B and C. Each page has an **Export PDF** button.

Glossary — every term in one line

- **Spread / gap** — the price difference between two related contracts, traded as one bet.
- **WTI / Brent** — American / global crude oil.
- **Calendar spread (M1-M2)** — same oil, two different delivery months.
- **Butterfly / fly** — a 3-month combination testing whether the middle month is out of line.
- **Brent-WTI** — the gap between the two crudes.
- **Fair value / "normal"** — what a spread *should* be worth; the reference for rich/cheap.
- **Rolling fair value** — fair value = the average of the spread's own recent prices (price-derived).
- **Fundamental fair value** — fair value predicted from real-world causes (inventories, dollar, VIX...); the Phase-2 model.
- **Rich / cheap** — a spread stretched unusually wide (rich) or narrow (cheap) vs fair value.
- **Wiggle / standard deviation** — the spread's typical jiggle; the unit we measure stretch in.
- **Z-score / stretch** — how many wiggles from normal we are right now.
- **Fade** — bet against the recent move (sell the rich, buy the cheap).
- **Long / Short** — a buy (profits if it rises) / a sell (profits if it falls).
- **Mean reversion** — the tendency of a stretched spread to snap back to normal.

- **Take profit / target** — closing a winning trade once it has reverted through fair to ~1.5 wiggles of overshoot.
 - **Overshoot** — a snapping-back spread usually shoots a bit past fair before settling; riding it captures more of the swing.
 - **Stop loss** — the pre-set point where we admit we're wrong and cut.
 - **Time stop** — closing a trade that hasn't resolved within a set time.
 - **Walk-forward / out-of-sample** — only ever training on the past, so the test never sees the answer; the honest way.
 - **Look-ahead** — the cheating-by-accident of using future information; what walk-forward prevents.
 - **Contract roll** — when the nearest contract expires and the "continuous" spread shifts to the next month, causing a jump.
 - **Backtest** — a trade-by-trade simulation of the strategy on historical data.
 - **Gross vs net** — results before costs (gross) vs after costs (net); net is the tradeable truth.
 - **Cost gate** — only take a trade when its expected dollar move clears $\sim 2\times$ the trading cost; filters out cent-sized churn (volatility-adaptive).
 - **Turnover** — how often a strategy trades; high turnover multiplies cost and is what sank the pre-gate intraday net.
 - **Universe pruning** — trading only the structures with a persistent edge after costs in *both* halves of history.
 - **Profit factor** — dollars won $\div$ dollars lost (above 1 = profitable).
 - **Expectancy** — the average profit per trade.
 - **Win rate** — the % of trades that made money.
 - **Drawdown** — the worst peak-to-valley fall in the running total; the "pain" measure.
 - **Sharpe (per-trade)** — average profit $\div$ how bumpy the profits are; consistency.
 - **t-statistic** — how many "standard errors" the average trade is above zero; >2 means the edge is unlikely to be luck.
 - **Monte-Carlo drawdown** — reshuffle the trade order thousands of times to see how bad the worst dip could plausibly have been.
 - **Walk-forward selection** — re-choosing which structures to trade each year using only the past, to prove the choice isn't hindsight.
 - **Regime** — the market's current "mood" (e.g. Tight · High Vol), by inventory $\times$ volatility.
 - **Slippage** — the small cost paid on every buy and sell (the bid–ask).
 - **Contract / unit** — 1,000 barrels; a \$1/barrel spread move = \$1,000.
-

End of report. Three backtests, one honest conclusion: there are two tradeable edges — the slow, fundamentals-based daily strategy (cost-robust by design), and the fast intraday strategy once it is cost-disciplined (the cost gate, pruning and overshoot exit lift its net from \$50k to \$1.98M — on a higher gross, profitable every year — and it holds up out-of-sample). The lesson that the gross numbers hide is also the fix: measure net, then trade only when the edge clears the cost, and capture the whole move. Everything here is gross-and-net, walk-forward where it matters, independently audited, and stated with its limits.